

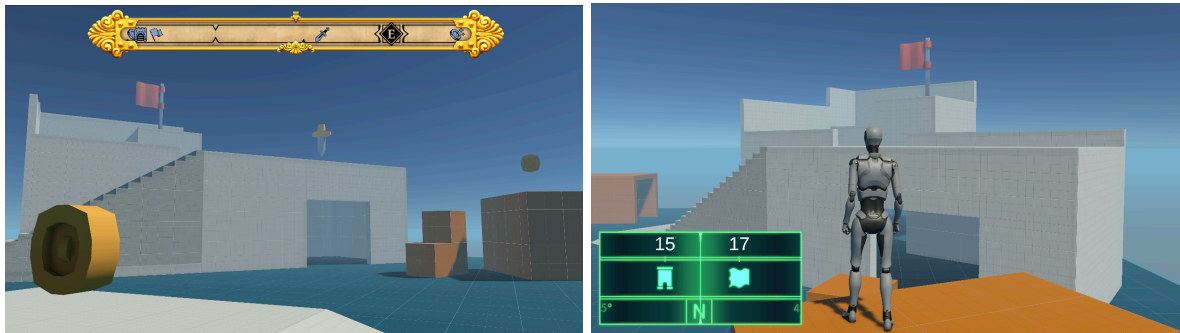
Bethesda-style Compass

General Description

This compass works just like the ones in Bethesda games, such as Skyrim or Fallout 3. It gives the player the direction towards certain game objects in the scene: characters, items, locations, etc.



The compass works in both first and third person modes. In the former, the camera and the character are facing the same direction, while that is not the case in the latter.. This is why the compass direction is based on that of the camera, not the one the main character is facing.



Features

Background with cardinal directions

You can add a background image of any length. It will update the current direction the camera is facing. North corresponds to the forward direction in world space.

One full rotation will cover the entire image. So the shorter the image, the faster it will scroll.

This feature is managed by the `CompassBackgroundController` script.

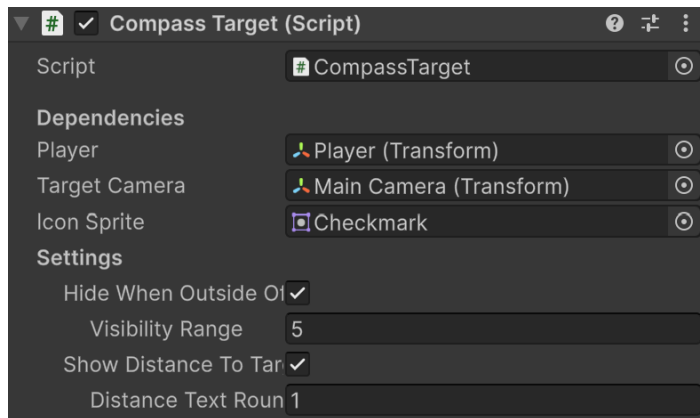
Icons on the compass

The `CompassIconController` adds, removes and defines the position of the icons on the compass. Each icon corresponds to a game object in the scene carrying the `CompassTarget` script. If an object is outside of the compass's view, then it will be displayed on its closest edge.

Icon targets

The CompassTarget class give you control over the customization of the icon:

- A sprite
- A maximum visibility range
- The option to hide or show the icon when the target is beyond this range
- The option to show the distance to the target on top of the icon
- The distance number decimal precision



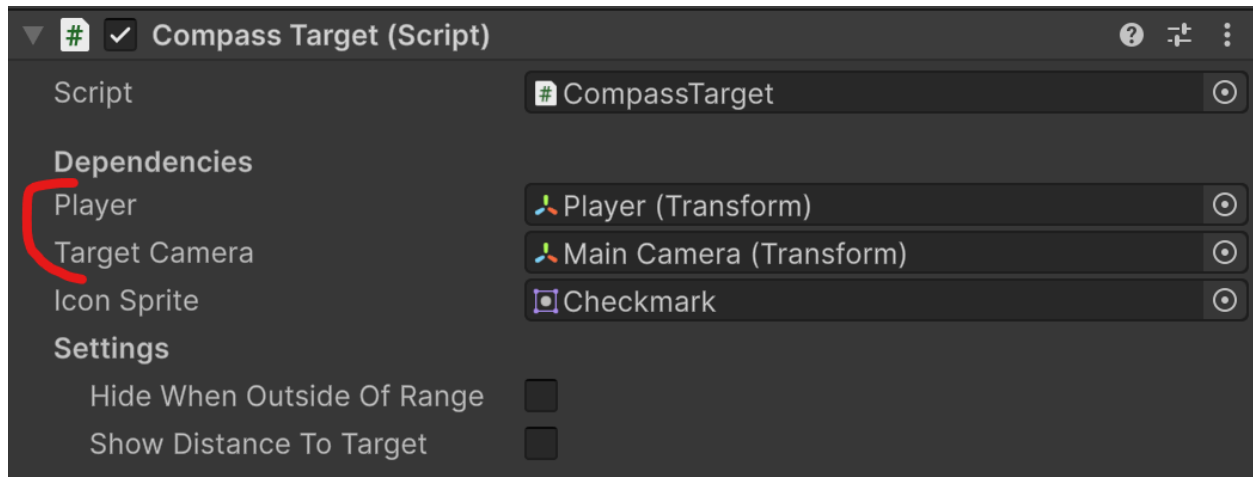
The compass can be placed anywhere on the screen and still display the icons at their right positions.



Adding/removing icons

Icons are added in one of two ways.

Option 1: When a scene is loaded. In that case you need to drag and drop the references to the player and camera.



Option 2: Manually adding the icon at runtime by using the public Setup method from the CompassTarget class:

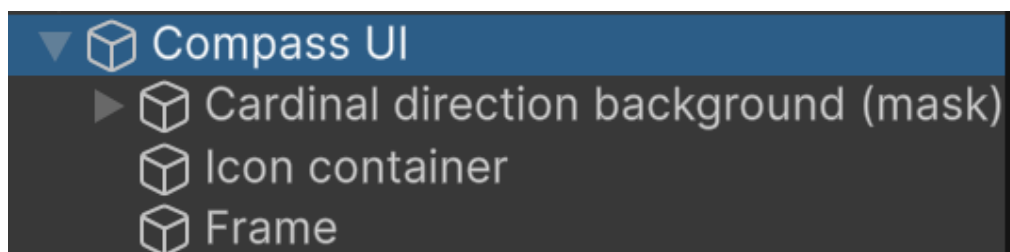
```
public class CompassTarget : MonoBehaviour
{
    0 references
    public void Setup(Transform player, Transform cameraTransform)
    {
        this.player = player;
        this.targetCamera = cameraTransform;
    }
}
```

A good example of this would be an enemy spawner, where you would pass the references of the camera and the player for each new enemy instantiated.

Icons are removed when the associated target game objects are either disabled or destroyed.

Customizing the compass

The compass is made of 3 distinct parts: the frame, the background and the icon container. The frame is just an image, there is no logic attached to it. The background controls the scrollable image with the cardinal directions. Finally, the icon container defines the area where the icons are displayed on the compass.

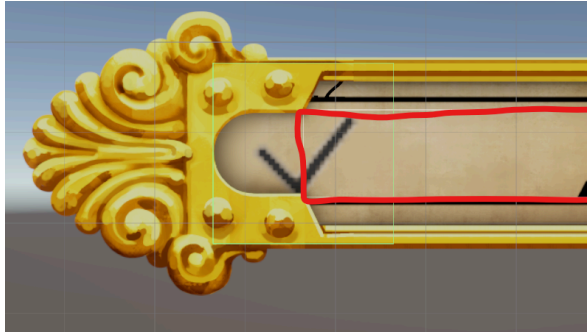


All 3 elements are customizable. For instance, if you want to create a compass of different dimensions for your game, you can provide a different frame and then adjust the size of the background and icon container to match the new layout.

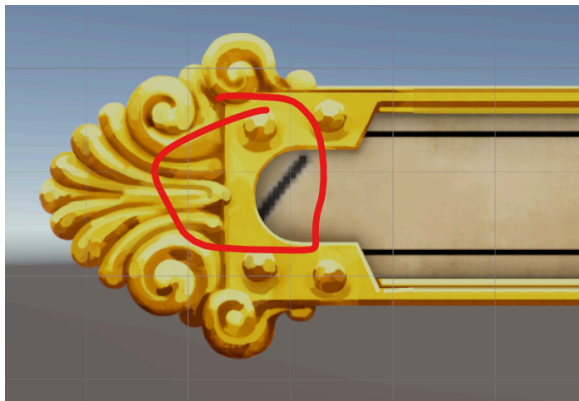
One important detail to note is that the background and the icon container are independent, and thus don't need to be of the same dimensions. In this example, I made the icon container narrower than the background, so that the icon would not be partially hidden at the edges of the "viewport" of the compass.



In red are the boundaries of the icon container:



If I were to make the icon container as wide as the background, the icon would be hidden, like so:



Credits

Programming

Willy Mercier

- Portfolio -> <https://www.willymercier-portfolio.com/>

Art

Theodore Matthews

- Instagram -> @theodoremattthews
- BlueSky -> @tiwazmannaz